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Pipeline Flow Rate Real-Time Monitoring Based on Optical FBG Fiber Acceleration Sensor

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Abstract

Non-invasive flow sensors will not interfere with the flow characteristics in the pipeline, which is the main research direction of downhole permanent flow sensors at present. In the extremely harsh environment of high temperature, high pressure, corrosion, and so on, the optical fiber sensor has higher reliability than the traditional electric sensor. The standard deviation of pipeline vibration is directly proportional to the wall pressure fluctuation caused by flow turbulence. Therefore, based on the fiber grating sensing principle and sensor geometry optimization method, a high-sensitivity fiber optic accelerator sensor was designed in this paper to sense the wall pressure fluctuation for non-invasive pipeline flow measurement. An experimental platform was set up to carry out relevant experiments. The standard deviation of vibration and the experimental data of flow rate were fitted by exponential fitting and quadratic fitting methods, and the data indicate a fit of up to 98%. In the range of flow rate fitting error of $\pm 5\%$, the exponential fitting method was better, and the lowest measured flow rate was 0.37 m/s. The experiment proves that the accelerator sensor can realize the non-invasive real-time measurement of underground pipeline wells.

Key words: Underground pipeline, Flow monitoring, Pipe wall pressure, Fiber bragg grating acceleration sensor

With the continuous development of oilfield development technology, the construction of fully automated "intelligent oilfields" is gradually becoming a trend in the current development of the petroleum industry, and a downhole permanent sensor is an indispensable downhole measuring instrument^[1] to realize this goal. In the extremely harsh environment of high temperature, high pressure, corrosion, strong electromagnetic interference, and so on, the reliability and long-term stability of traditional electrical sensors are extremely limited^[2]. Fiber Bragg grating sensor has become the first choice for downhole testing sensing instruments because of its small size, lightweight, and strong ability to resist extreme harsh environments such as high temperature and high pressure^[3].

The downhole flow sensor can obtain the flow value at the specific depth of the well, which is of great value for analyzing the current situation of the reservoir, guiding production and development, and formulating an oil recovery plan. Due to the complex composition of the fluid in the pay zone, it usually carries sand particles. At present, there are dozens of flowmeters used to achieve flow measurement on the

surface, but most are not suitable for long-term use^[4] in the downhole. Non-invasive flowmeter does not contact with the inside of the pipeline when in use, will not interfere with the internal flow characteristics of the pipeline, and has a greater guarantee of long-term reliability.

The non-invasive flow monitoring is based on the principle of using flow-induced vibration signal for flow rate measurement proposed by Pittard M T et al. It is the first time to prove that the standard deviation of the noise of pipeline vibration signal has a linear relationship with flow rate^[5]. By experimentally testing a pipe of known length, diameter, and material, Giuseppe Dinardo et al. confirmed that the amplitude of the first harmonic of the vibration signal is proportional to the flow rate. Daniel Pontes Lannes et al^[6], installed a piezoelectric sensor on the inner surface of the pipe to detect turbulent pressure fluctuations and experimentally verified that the pressure fluctuations at the bend were higher^[7]. Shujuan Li et al. designed a fiber laser vibration sensor with a hinged structure and obtained the vibration intensity of the pipeline through fiber interference demodulation technology, and then calculated the flow rate^[8]. Tongda Li et al. also proposed a non-invasive flow sensing system based on DAS technology, and numerically analyzed wall pressure fluctuations under different flow rates to obtain flow rate values^[9].

In this paper, a method that can be used for real-time monitoring of pipeline flow was implemented. fiber Bragg grating acceleration sensor was used to sense the vibration signal of the pipe wall and demodulate the wavelength signal of fiber grating in real-time. The change in wavelength data reflected the size of the pipeline vibration signal. The wavelength value was input into the host computer in real-time for processing, and its fitting mode, fitting data amount, and fitting error were studied. The standard flow fitting measurement model was made, and the current real-time flow rate was obtained, to achieve a long-term, stable, and reliable pipeline flow monitoring effect.

Turbulence intensity flow measurement principle

The flow of fluid in the pipeline produces turbulence, and when it is close to the wall of the pipeline, kinetic energy will be generated. Most of it is converted into potential energy in the form of pressure. The vibration in the pipeline is caused by this energy conversion. From Evans' derivation, it can be seen that pressure fluctuation is^[10] proportional to velocity fluctuation:

$$p' \propto \overline{u'v'} \quad (1)$$

Where: p' is the pressure fluctuation, u' is the fluid axial speed fluctuation, v' is the fluid radial speed fluctuation.

The pipeline filled with liquid was simplified into a one-dimensional beam model, and combined with the generosity equation of the beam, the pipeline acceleration and pressure fluctuation were established as follows:

$$\frac{\partial^2 y}{\partial t^2} = -\frac{\theta g}{A\gamma} \frac{\partial^4 y}{\partial x^4} = \frac{\partial^4 y}{\partial x^4} \propto p(x) \quad (2)$$

In the formula: A is the cross-sectional area of the beam, γ is the specific gravity of the beam, g is the acceleration of gravity, EI is the bending stiffness.

For dynamically similar flows, the ratio of flow fluctuation to average flow is constant, and this relationship was defined as "turbulence intensity"^[11]:

$$\frac{\sqrt{m}}{\bar{u}} = \frac{\frac{1}{N} \sum_{i=1}^N [u_i(t) - \bar{u}]^2}{\bar{u}^2} = C \quad (3)$$

Where: $\bar{u}$ is the average value of fluid velocity, $u_i(t)$ is the instantaneous value of fluid velocity, N represents the number of data points, and C is the turbulence intensity.

The relationship between the standard deviation of pipeline vibration and the average flow rate was obtained by conversion as follows:

$$\frac{1}{N-1} \sum_{i=1}^N [u_i(t) - \bar{u}]^2 = \frac{1}{N-1} u^2 = Ku^2 \quad (4)$$

Where: K indicates the proportion coefficient.

From equation (5), we can see that the variance of the flow fluctuation in the pipe is in a quadratic relationship with the mean flow u . The standard deviation of the signal noise was fitted to the flow velocity using a quadratic linear model. A quadratic linear fit was performed to determine the relationship between the standard deviation of the signal noise and the flow velocity.

$$STD = k_2 \bar{u}^2 + k_1 \bar{u} + k_0 \quad (5)$$

Where: STD represents the standard deviation of signal noise and k_0 , k_1 , k_2 is the quadratic fitting coefficient.

The approximation of the function was carried out according to the polynomial. In general, the higher the fitting frequency of the polynomial, the more data is needed and the more accurate the fitting result is. Therefore, the quadratic fitting was further optimized into the form of polynomial fitting, and an exponential fitting formula is proposed:

$$STD = A \left(1 + \frac{\bar{u}}{c} + \frac{\bar{u}^2}{2c^2} + \frac{\bar{u}^3}{6c^3} + \dots \right) \quad (6)$$

Where: A is the coefficient of the constant term of the exponential fitting curve, and c is the coefficient of exponential fitting.

$$Ae^{\frac{\bar{u}}{c}} \xrightarrow{\text{Taylor}} A \left(1 + \frac{\bar{u}}{c} + \frac{\bar{u}^2}{2c^2} + \frac{\bar{u}^3}{6c^3} + \dots \right) \quad (7)$$

Where: $\text{Taylor}(Ae)$ represents the expansion of Taylor series for the independent variable Ae . The formula contains the first term and the second term in the original quadratic fitting formula, which can more accurately reflect the relationship between the actual flow rate transformation and standard deviation. This paper is based on this basis to calculate and monitor the real-time flow rate of the pipeline.

Fiber Bragg grating acceleration sensor

Measurement principle of sensor

The fiber bragg grating acceleration sensor is used to measure the vibration signal as shown in Fig.1(a). The part of the sensor that feels the vibration of the pipeline is the core body as shown in Fig. 1(b). When the water flow in the pipe produces vibration, the mass block will vibrate up and down along the base, and the fiber grating will be stretched and compressed, resulting in the change of the grating wavelength, and the wavelength signal size reflects the vibration amplitude of the pipeline.

Optimization design of sensor

It is known that the natural frequency is the frequency generated by the free vibration of the object, which is only related to the inherent properties of the object (such as material, density, size, etc.). The pipeline is simplified into an ordinary beam model, and its natural frequency formula is^[14] as follows:

$$f_i = \frac{i^2 \pi}{2L^2} \sqrt{\frac{E}{\rho A_q}} \quad (8)$$

Where: L is the length of the beam, A_q is the cross-sectional area of the beam, E is the density of the beam, and ρ is the elastic modulus.

In this paper, the pipe was made of 304 stainless steel, its elastic modulus is 193GPa, density is 7.85g/cm³, Poisson's ratio is 0.247, and the first order natural frequency of the pipe is calculated to be 26.95Hz. Because when the vibration is generated by the interaction between the water flow and the pipe wall, the vibration signal measured when the pipe resonates is the largest^[12] amplitude. For pipes of different materials, it is

necessary to calculate the first order natural frequency. Finally, the acceleration sensor was optimized to make it have higher sensitivity when the pipeline vibration reached the natural frequency.

The hinge size is the main factor^[13] affecting the sensitivity and natural frequency of the acceleration sensor. The sensor with higher sensitivity was obtained by single objective optimization through simulation, and the resulting parameters are shown in Table 1.

Table 1—Sensor core size parameters

parameter	quantity	value(mm)
d	semi-major axis	4
b	semi-minor axis	7
t	Thickness of hinge	0.8
l	Width of FBG	5
h	Height of mass	24
e	Width of mass	23
m	Thickness of mass	8

The natural frequency and sensitivity of the sensor were calibrated by testing on the vibration table. Firstly, the acceleration value of 1g was loaded, the frequency was set to 30Hz-340Hz, the step length is 10Hz, and the amplitude-frequency curve was obtained as shown in Fig.2(a); Then the acceleration value of 1g to 4g for the frequency of 30Hz, 40Hz, 50Hz, 60Hz and 70Hz were loaded respectively, with the step length of 1g, and the sensitivity calibration curve was gotten as shown in Fig.2(b). Finally, the frequency range of the sensor was 0-100Hz, and the sensitivity was 176.29pm/g at 30Hz.

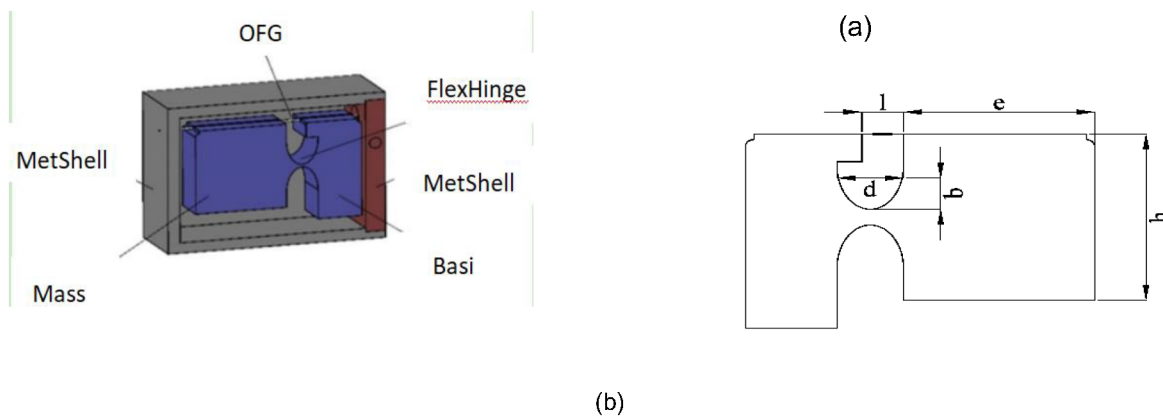
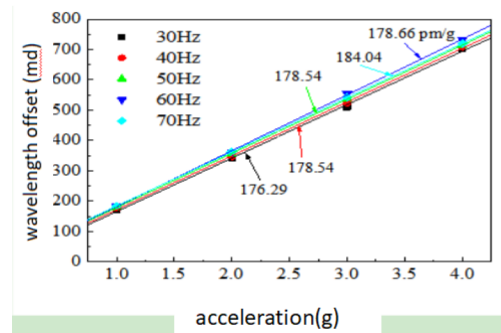
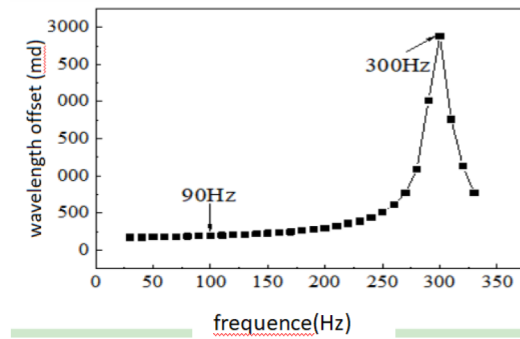


Figure 1—Sensor model diagram (a) Structure diagram (b) Core size parameter diagram



(a)



(b)

Figure 2—Frequency-sensitivity curve (a) amplitude-frequency curve (b) Sensitivity calibration curve

Curve fitting calibration

For acceleration sensors of different sizes, due to the difference in sensitivity and resonant frequency, the standard deviation and the fitting curve of the flow rate are also different, so the fitting curve needs to be calibrated in the first step of the experiment. The specific method is as follows: adjusting the frequency of the inverter from large to small at the frequency step of 3Hz, recording the vibration signal under the corresponding flow rate of each frequency and the flow rate displayed by the turbine flowmeter, and fitting the standard fitting curve of the vibration signal with the flow rate.

Fig. 3. shows the different calibration curves fitted before and after the size optimization of the sensor. The measurement accuracy of the sensor after the size optimization is nearly twice that of the sensor before optimization.

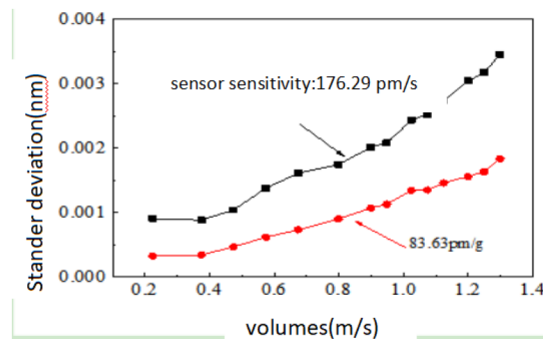
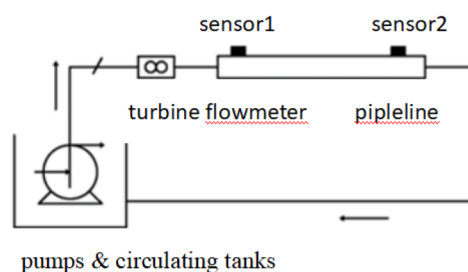


Figure 3—Fitting calibration curves

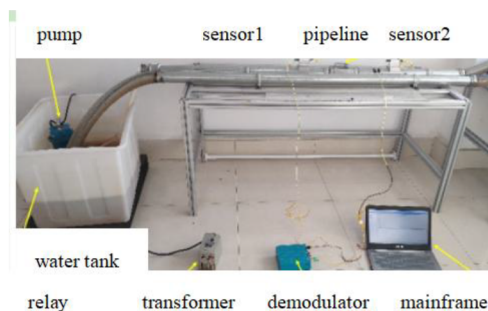
Experiment

Experimental Setup

The pipeline water circulation system built in this experiment is shown in Fig. 4., which is composed of a relay, frequency converter, circulating water tank, pump, turbine flowmeter, and test pipeline loop. The pipe is a steel pipe with a diameter of 50mm and a wall thickness of 1mm and a length of 180cm. The metering accuracy of the turbine flowmeter can reach $\pm 0.5\%$. Turn on the pump to make the pipeline circuit full of water, and change the flow rate by controlling the pump speed through the inverter. The frequency change range of the inverter is 15-55Hz, and the flow rate of the pipeline can be achieved in the range of 0.24-1.27m/s. At low flow rates, the influence of environmental noise is too large. Foam pads are placed under the water tank to remove the vibration effect of the water pump on the experimental pipeline support. The experimental data show that the vibration of the pump is the main factor affecting the vibration result of the pipeline.



(a)



(b)

Figure 4—Experimental flow chart (a) Water circulation loop (b) physical diagram

Pretreatment for denoising

The vibration signal collected by the acceleration sensor mainly includes two parts, one is the tube wall vibration induced by fluid flow vibration, and the other is the mechanical noise generated by the pump motor and environmental elements. The noise signal and the flow induced vibration signal are randomly distributed, and the lower the flow velocity, the larger the signal-to-noise ratio of the vibration signal. In order to improve the measurement accuracy of the low flow velocity, the original signal collected needs to be denoised.

For processing based on Matlab, the noise signal was randomly distributed and the mean value is constant. Figure 5 shows the effect of vibration signal after five times of accumulation and denoising at the flow rate of 0.89m/s. The signal envelope was similar to the original signal, and the noise attenuation degree at 300Hz was large, which was effectively filtered out.

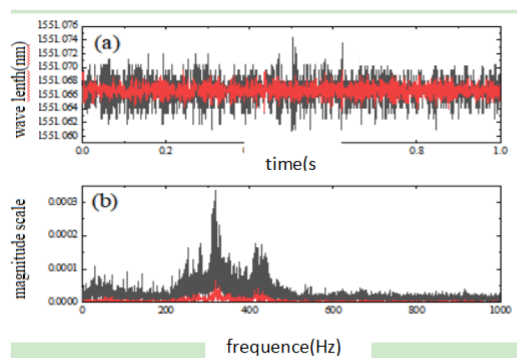


Figure 5—Signal and amplitude-frequency curve after cumulative denoising

Quadratic fitting and exponential fitting

Draw the standard curve fit to be used as the pipeline flow prediction model. To obtain the flow accurately, the effect of the fit needs to be studied. In known polynomial fitting, the higher the fitting times, the closer to the actual given data points. The vibration signal of five seconds (10000 points) collected by the experiment was selected for two to six times polynomial fitting, and the fitting results are shown in Figure 6. The fitting degree of quadratic fitting is up to 99.30%, while that of sixth-degree polynomial fitting is up to 99.72%. Therefore, in a certain range, higher fitting times can further improve the fitting accuracy and provide a data basis for exponential fitting.

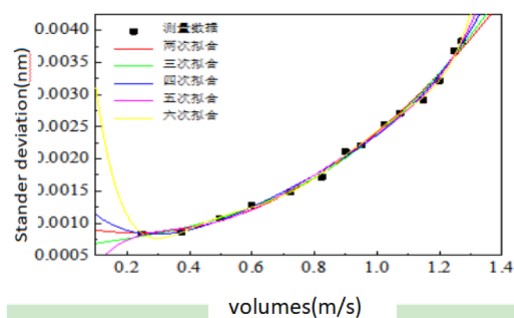
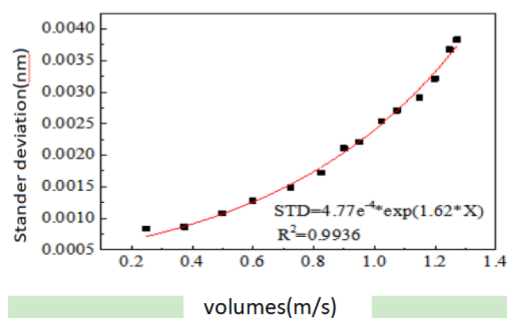
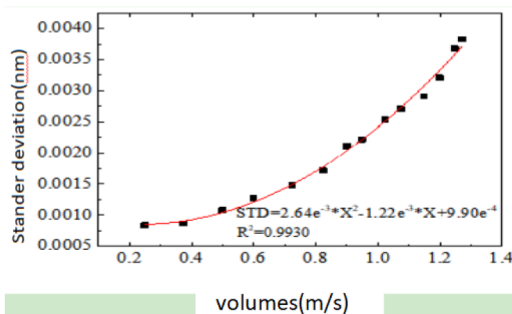


Figure 6—Polynomial fitting of order 2-6

The following two flow measurement models of quadratic fitting and exponential fitting were tested, vibration signal data were collected under the same test conditions, and the standard deviation – flow rate fitting curve was drawn. As can be seen from FIG. 7, under all the measured flow rates, the fitting curve was closely fitted to the discrete points of the original data, the fitting degree (R-squares) was above 99%, and the exponential fitting degree was slightly higher than the quadratic fitting degree.



(a)



(b)

Figure 7—Fitting curve (a) Quadratic fitting (b) exponential fitting

The relative errors of the two fitting methods are plotted in Fig. 8. It can be seen that within the range of $\pm 5\%$ fitting error, exponential fitting can correctly reproduce a lower flow rate, as low as 0.37m/s, than quadratic fitting.

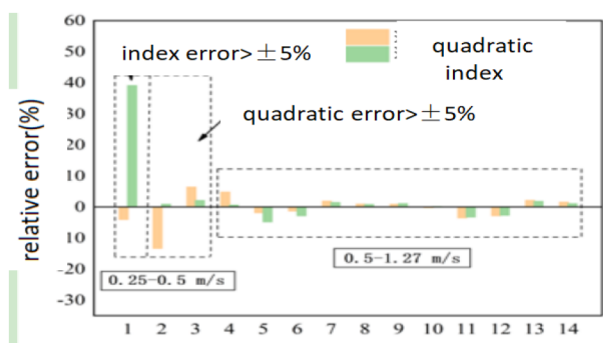


Figure 8—Accuracy verification

Fig.9. further tests the repeatability of inverse velocity with exponential fitting. The fitting degree of four repeated experiments is 98.54%, 98.70%, 98.42%, and 98.91%, respectively.

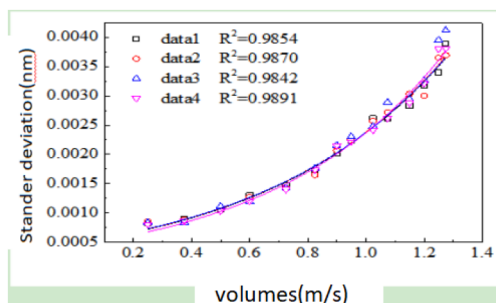


Figure 9—Repeatability verification

Real-time flow rate monitoring

When the sampling frequency of the demodulation instrument is 2000Hz, based on the same sensor, 0.5s was taken as a step, and the selected time length was successively reduced for exponential fitting. The results are shown in Fig. 10. As the number of sampling points decreases, the fitting degree between the standard deviation and flow rate curve became smaller, which was reflected in the decrease of standard deviation at a higher flow rate. When the number of points was 2000, the fit degree could reach 98%, which provided a basis for the real-time monitoring of the system.

The time-frequency domain feature extraction of vibration signals was carried out, and the corresponding relationship between characteristic factors and velocity was drawn respectively. Finally, eight characteristic values, namely standard deviation, mean square value, root mean value, waveform factor, peak factor, pulse factor, margin factor and singular spectrum entropy, were selected for neural network traffic prediction. 2000 vibration signal points were selected under each velocity of the collected signal, and the characteristic values were calculated as input to predict the flow rate. Fig 11. shows the prediction results, which can be used as a data reference for flow measurement by the fitting method. When the flow velocity is greater than 1.0m /s, the prediction error increases sharply, indicating that this method is more suitable for low flow velocity prediction.

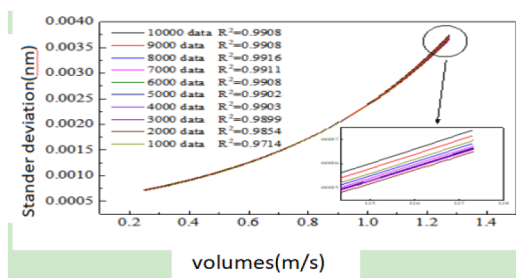


Figure 10—Point fitting verification

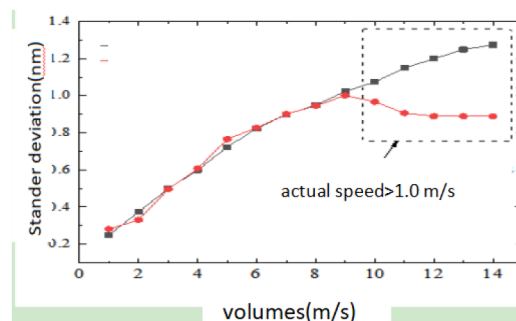


Figure 11—LSTM neural network flow prediction

Conclusion

Based on the fiber grating sensing principle and sensor geometry optimization method, a fiber accelerator sensor with a sensitivity of 176.29pm/g is designed in this paper to sense the wall pressure fluctuation and realize non-invasive pipeline flow measurement. Moreover, an experimental platform was set up to carry out relevant experiments. At the same time, exponential fitting and quadratic fitting methods were used to fit the experimental data of vibration standard deviation and flow rate, and the fitting degree was as high as 98%. Within the fitting error range of $\pm 5\%$, the exponential fitting method was better, and the lowest flow rate that could be measured was 0.37m /s. The experimental results show that the sensor can better realize the non-intrusive real-time measurement of underground pipelines.

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